

Important Tests for Econ 104

Midterm

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Heteroskedasticity Tests: BP Test

1. $y_i = E(y_i) + e_i = \beta_1 + \beta_2 x_{i2} + \dots + \beta_K x_{iK} + e_i$ Do your main regression
2. $\hat{e}_i^2 = \alpha_1 + \alpha_2 z_{i2} + \dots + \alpha_S z_{iS} + v_i$ Regress your residuals² (sample variance) on variables you believe to be correlated w/ var
3. Null hypothesis – homoskedasticity: NR^2 follows Chi Sq (df = # of z)
 - $Pval < .05$ or $NR^2 > \text{Chi Sq critical value}$, Reject Null hypothesis of homoskedasticity and conclude that there is heteroskedasticity
 - Use R^2 of model under 2

Heteroskedasticity Test: White Test

- Exactly the same as BP, except you don't choose the z
 - You just use all of the x, the squared x, and potentially their cross product (you don't need to impose where the variance is coming from)
- Model: $E(y) = \beta_1 + \beta_2 x_2 + \beta_3 x_3$
- Z's: $z_2 = x_2, z_3 = x_3, z_4 = x_2^2, z_5 = x_3^2$ maybe $z_6 = x_2 x_3$
- Then same as BP: null homoscedastic $\sim$ chi sq

Heteroskedasticity Test: Goldfeld-Quandt Test

1. Get main model: $\widehat{WAGE} = \frac{-9.914}{(1.08)} + 1.234 \frac{EDUC}{(0.070)} + 0.133 \frac{EXPER}{(0.015)} + 1.524 \frac{METRO}{(0.431)}$

2. Split Observations into 2 groups:

$$\begin{aligned} WAGE_{Mi} &= \beta_{M1} + \beta_{M2} EDUC_{Mi} + \beta_{M3} EXPER_{Mi} + \beta_{M4} METRO_{Mi} + e_{Mi} \\ \text{for } i = 1, \dots, N_M \end{aligned}$$

$$\begin{aligned} WAGE_{Ri} &= \beta_{R1} + \beta_{R2} EDUC_{Ri} + \beta_{R3} EXPER_{Ri} + \beta_{R4} METRO_{Ri} + e_{Ri} \\ \text{for } i = 1, \dots, N_R \end{aligned}$$

3. H0: Homoskedasticity: $\sigma_M^2 = \sigma_R^2$

- Under H0: $F = \frac{\hat{\sigma}_M^2}{\hat{\sigma}_R^2}$
- 2-sided test: Reject H0 if $F < F_{LC}$ or $F > F_{UC}$. where LC = left crit val
- If reject H0, Heteroskedasticity: 2 groups have different variance

$SSE_1/SSE_2 \sim F$ under H0

- Degrees of freedom are n-k

Serial Correlation Test (Heteroskedasticity in Times Series): Lagrange Multiplier Test

- Very Similar to BP test, but for time series

1. Get main model: $y_t = b_1 + b_2x_t + \hat{e}_t$

2. Impose model with serial correlation on residual: $e_t = \rho e_{t-1} + v_t$

3. New model becomes: $y_t = \beta_1 + \beta_2x_t + \rho\hat{e}_{t-1} + v_t$

4. Rearranging: $\hat{e}_t = \gamma_1 + \gamma_2x_t + \rho\hat{e}_{t-1} + v_t$

5. Under H0: Homoskedasticity $\rightarrow \rho = 0 \rightarrow TR^2 \sim \chi^2$

- Testing R^2 from 4
- T is how many time periods in sample

VAR Causality Test: Granger Causality Test

Granger| causality test H_0 : starts do not cause comps

Model 1: comps ~ Lags(comps, 1:4) + Lags(starts, 1:4)

Model 2: comps ~ Lags(comps, 1:4)

	Res.Df	Df	F	Pr(>F)
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1	329			
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2	333	-4	32.125	< 2.2e-16 ***
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A: Reject $H_0 \rightarrow$ Yes!

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Looks at whether the First Model is a better predictor than the second model. If so, the included predictors are “good” predictors

IV Test: Weak Instrument (F-Test)

```
## Call:
## ivreg(formula = log(wage) ~ educ + exper + I(exper^2) | exper +
##       I(exper^2) + mothereduc + fathereduc, data = mroz1, singular.ok = TRUE)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.0986 -0.3196  0.0551  0.3689  2.3493
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.0481003  0.4003281   0.120  0.90442
## educ         0.0613966  0.0314367   1.953  0.05147 .
## exper        0.0441704  0.0134325   3.288  0.00109 **
## I(exper^2)   -0.0008990  0.0004017  -2.238  0.02574 *
##
## Diagnostic tests:
##              df1 df2 statistic p-value
## Weak instruments  2 423    55.400 <2e-16 ***
## Wu-Hausman       1 423     2.793  0.0954 .
## Sargan           1  NA     0.378  0.5386
```

First Stage:

$$\widehat{EDUC} = \begin{matrix} 9.7751 \\ (se) \end{matrix} + \begin{matrix} +0.0489 \\ (0.0427) \end{matrix} EXPER - \begin{matrix} -0.0013 \\ (0.0012) \end{matrix} EXPER^2 + \begin{matrix} +0.2677 \\ (0.0311) \end{matrix} MOTHEREDUC$$

Second Stage:

$$\ln(\widehat{WAGE}) = \begin{matrix} 0.1982 \\ (se) \end{matrix} + \begin{matrix} +0.0493 \\ (0.0374) \end{matrix} \widehat{EDUC} + \begin{matrix} +0.0449 \\ (0.0136) \end{matrix} EXPER - \begin{matrix} -0.0009 \\ (0.0004) \end{matrix} EXPER^2$$

1. Run our 2SLS above
2. Look at First stage, we are checking whether our instrument (MOTHEREDUC) influences EDUC
 - May be more than 1 instrument
3. Under H0, instrument doesn't influence endogenous predictor (EDUC)
 - Under H0 ~ F (jointly test whether coefficients on instruments are = 0)
 - Rule of Thumb: F-statistic > 10 -> Strong instrument (reject H0), F-stat < 10 -> weak instrument

IV Test: Wu-Hausman

```
## Call:
## ivreg(formula = log(wage) ~ educ + exper + I(exper^2) | exper +
##       I(exper^2) + mothereduc + fathereduc, data = mroz1, singular.ok = TRUE)
##
## Residuals:
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```

1. Run our 2SLS above
 2. Get Residuals from second stage
 3. Include Residuals in First Stage:
 - Under H_0 : $\delta = 0$ Both OLS and IV are consistent
 - Under H_1 : OLS is not consistent, while IV is consistent
- Overall, testing whether we need to use IV to get a consistent estimator or if OLS would work
 - Assumes our instrument is exogenous (assumes we have a “good” instrument)

Second Stage:

$$y = \beta_1 + \beta_2 x + e$$

First Stage:

$$x = \gamma_1 + \theta_1 z_1 + \theta_2 z_2 + v$$

Get Residuals:

$$\hat{v} = x - \hat{\gamma}_1 + \hat{\theta}_1 z_1 + \hat{\theta}_2 z_2$$

Include Residuals in First Stage:

$$y = \beta_1 + \beta_2 x + \delta \hat{v} + e$$

$$H_0 : \delta = 0 \quad \text{i.e., no correlation between } x \text{ and } e$$

$$H_1 : \delta \neq 0 \quad \text{i.e., correlation between } x \text{ and } e$$

IV Test: Sargan

```
## Call:
## ivreg(formula = log(wage) ~ educ + exper + I(exper^2) | exper +
##       I(exper^2) + mothereduc + fathereduc, data = mroz1, singular.ok = TRUE)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.0986 -0.3196  0.0551  0.3689  2.3493
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```

Second Stage:

$$y = \beta_1 + \beta_2 x + e$$

First Stage 1:

$$x = \gamma_1 + \theta_1 z_1 + v$$

First Stage 2:

$$x = \gamma_1 + \theta_1 z_1 + \theta_2 z_2 + v$$

- Idea: We know z_1 is a good instrument but unsure about z_2 (not sure whether to include z_2)
1. Run our 2SLS above
 2. Under H_0 : Extra instrument is good (the second stage is not over-identified)
 - H_1 : Extra instrument is bad (second stage is over-identified)